

Analytical MCQ Assessment

High-level cognitive evaluation paper.

SECTION A: Multiple Choice Questions

Q1. A network administrator wants to block incoming traffic based solely on source IP and port number without tracking the state of active connections. Which type of firewall is most appropriate? (1 Marks)

- A. Stateful Inspection Firewall
- B. Packet Filtering Firewall
- C. Application-Level Gateway
- D. Next-Generation Firewall

Q2. Which type of firewall operates at the Session layer of the OSI model to monitor TCP handshaking and determine if a requested session is legitimate? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-Level Gateway
- C. Application-Level Gateway
- D. Stateful Inspection Firewall

Q3. An organization requires a firewall that can inspect the actual payload of HTTP requests to block specific SQL injection patterns. Which traditional firewall type is best suited for this? (1 Marks)

- A. Packet Filtering Firewall
- B. Circuit-Level Gateway
- C. Application-Level Gateway
- D. Stateful Inspection Firewall

Q4. How does a stateful inspection firewall determine whether to allow an incoming packet that does not match an explicit inbound rule? (1 Marks)

- A. It automatically blocks all unsolicited incoming packets regardless of context.
- B. It checks if the packet is part of an established outbound connection recorded in its state table.
- C. It performs deep packet inspection on the payload to guess the intent.
- D. It forwards the packet to a proxy server for further analysis.

Q5. What primary capability distinguishes a Next-Generation Firewall (NGFW) from a traditional stateful inspection firewall? (1 Marks)

- A. Ability to filter traffic based on source and destination IP addresses.
- B. Integration of deep packet inspection with application awareness and intrusion prevention systems.
- C. Support for Virtual Private Network (VPN) termination.
- D. Operating at the Network layer of the OSI model.

Q6. A company needs to protect its web servers from Cross-Site Scripting (XSS) and SQL injection attacks. Which specialized firewall should be deployed? (1 Marks)

- A. Next-Generation Firewall (NGFW)
- B. Web Application Firewall (WAF)
- C. Circuit-Level Gateway
- D. Stateful Packet Filter

Q7. Which scenario best describes the deployment of a host-based firewall? (1 Marks)

- A. A hardware appliance protecting the perimeter of an enterprise network.
- B. A software application running on an individual endpoint to control its traffic.
- C. A cloud-based service filtering traffic for multiple branch offices.
- D. A virtual appliance routing traffic between different VLANs.

Q8. Why might a traditional packet filtering firewall fail to stop an attack that uses port 80 (HTTP)? (1 Marks)

- A. Packet filters cannot block port 80.
- B. Packet filters do not inspect the payload of the packet to detect malicious commands.
- C. Packet filters only inspect UDP traffic.
- D. Port 80 is automatically bypassed by stateful inspection rules.

Q9. Which firewall type is most closely associated with hiding internal IP addresses by acting as an intermediary for all client requests? (1 Marks)

- A. Packet Filtering Firewall
- B. Proxy Firewall
- C. Circuit-Level Gateway
- D. Stateful Inspection Firewall

Q10. What is a major performance trade-off when choosing a stateful inspection firewall over a stateless packet filtering firewall? (1 Marks)

- A. Stateful firewalls require more memory and CPU to maintain the connection state table.
- B. Stateful firewalls are less secure because they do not inspect packet headers.
- C. Stateful firewalls cannot handle TCP traffic.
- D. Stateful firewalls require manual rule updates for every new connection.

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

MCQ Solutions & Rationale

Q1. Correct Answer: B

Rationale: Packet filtering firewalls inspect packets individually based on header information without maintaining connection state.

Q2. Correct Answer: B

Rationale: Circuit-level gateways operate at the Session layer to verify the TCP three-way handshake before establishing a connection.

Q3. Correct Answer: C

Rationale: Application-Level Gateways inspect the application layer payload to detect and block malicious content.

Q4. Correct Answer: B

Rationale: Stateful firewalls allow inbound traffic that belongs to an established outbound session recorded in their state table.

Q5. Correct Answer: B

Rationale: NGFWs combine traditional firewall capabilities with deep packet inspection and application-level control.

Q6. Correct Answer: B

Rationale: A WAF is specifically designed to protect web applications by filtering and monitoring HTTP traffic for application-layer exploits.

Q7. Correct Answer: B

Rationale: Host-based firewalls are software applications installed directly on individual endpoints to protect that specific host.

Q8. Correct Answer: B

Rationale: Traditional packet filters only inspect header information and cannot analyze the payload for malicious content.

Q9. Correct Answer: B

Rationale: A proxy firewall acts as an intermediary, establishing a separate connection to the destination and hiding the client's internal IP.

Q10. Correct Answer: A

Rationale: Maintaining a state table for active connections requires more system resources than stateless packet filtering.